NIOS Class 10 (212) Science & Technology — Syllabus Bifurcation & Exam Blueprint

NIOS Secondary (Class 10) Science & Technology (212) — Exam Prep Study Kit

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> **CRITICAL TIME-SAVING RULE FOR WORKING STUDENTS:**
> NIOS has officially bifurcated the syllabus into **TMA (40%)** and **Public Examination (60%)**.
> Out of 32 total textbook chapters, **13 chapters are TMA-only and will NEVER appear in the written exam!**
> You only need to study the **19 Public Examination chapters**.

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1. What to Study vs What to Completely Skip

Module & Subject Area	Total	❌ DO NOT STUDY (TMA Only - 40%)	✅ MUST STUDY (Public Exam - 60%)
Module 1: Measurement	1	*None*	• **L.1: Measurement in Science & Technology**
Module 2: Matter (Chemistry)	7	• L.2: Matter in our Surroundings • L.3: Atom and Molecules • L.4: Chemical Reaction & Equations	• **L.5: Atomic Structure** • **L.6: Periodic Classification of Elements** • **L.7: Chemical Bonding** • **L.8: Acids, Bases and Salts**
Module 3: Moving Things (Physics)	3	• L.9: Motion and its Description	• **L.10: Force and Motion** • **L.11: Gravitation**
Module 4: Energy (Physics)	7	• L.12: Sources of Energy • L.14: Thermal Energy • L.18: Sound and Communication	• **L.13: Work and Energy** • **L.15: Light Energy (Optics)** • **L.16: Electrical Energy** • **L.17: Magnetic Effect of Electric Current**
Module 5: The Living World (Biology)	7	• L.19: Classification of Living Organisms • L.20: History of Life on Earth • L.21: Building Blocks of Life – Cell & Tissues	• **L.22: Life Processes – I (Nutrition, Respiration, Circulation, Excretion)** • **L.23: Life Processes – II (Control & Coordination)** • **L.24: Life Processes – III (Reproduction)** • **L.25: Heredity**
Module 6: Natural Resources (Chemistry)	3	• L.26: Air and Water	• **L.27: Metals and Non-metals** • **L.28: Carbon and Its Compounds**
Module 7: Human & Environment (Ecology)	4	• L.31: Food Production • L.32: Health and Hygiene	• **L.29: Natural Environment** • **L.30: Human Impact on Environment**

Module & Subject Area	Total	❌ DO NOT STUDY (TMA Only - 40%)	✅ MUST STUDY (Public Exam - 60%)
TOTAL	**32 Lessons**	**13 Lessons SKIPPED (Save 40% Study Time!)**	**19 High-Yield Lessons ONLY**

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2. Question Paper Pattern (85 Marks, 2½ Hours)

The written paper contains **43 questions** totaling **85 Marks** (+15 marks practical = 100 marks total):

Section A: Objective Questions — 42 Marks (49.4% of the Paper!)

- **Q1 to Q17 (17 Questions × 1 Mark = 17 Marks)**: Multiple Choice Questions (MCQs). Formula definitions, SI units, symbols, basic concepts.
- **Q18 to Q27 (10 Questions × 2 Marks = 20 Marks)**: Objective Type (each question has two 1-mark subparts: Fill in the blanks, True/False, Match the column).
- **Q28 (1 Question × 5 Marks = 5 Marks)**: Context / Case Study question with five 1-mark subparts.

> **Scoring Insight**: Getting 35+ on these 42 objective marks guarantees you pass easily without breaking a sweat!

Section B: Subjective Questions — 43 Marks

- **Q29 to Q37 (9 Questions × 2 Marks = 18 Marks)**: Very Short Answer (VSA). Definitions, 2-line reasoning, short calculations.
- **Q38 to Q42 (5 Questions × 3 Marks = 15 Marks)**: Short Answer (SA). Chemical reactions, ray diagrams, 3 differences.
- **Q43 (1 Question × 5 Marks = 5 Marks)**: Long Answer (LA) with internal choice (Human physiology or Physics laws/numerical).
- **Q44 (1 Question × 5 Marks = 5 Marks)**: Long Answer (LA) with internal choice (Electricity/Light or Carbon/Salts).

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3. The 35-Mark "Sure-Shot" Topic List

If you master these 7 recurring topics, you will effortlessly score 35+ marks in Section B alone: 1. **Light & Optics (5 Marks)**: Mirror/Lens formula numerical ($\frac{1}{f} = \frac{1}{v} \pm \frac{1}{u}$) and Myopia/Hypermetropia defects with ray diagrams. 2. **Electricity (5 Marks)**: Ohm's law ($V = IR$), series vs parallel equivalent resistance, and electric power/energy billing numerical. 3. **Chemical Salts (4 Marks)**: Preparation, chemical equations, and uses of Bleaching Powder, Baking Soda, Washing Soda, and Plaster of Paris. 4. **Carbon Compounds (4 Marks)**: Saponification, cleansing action of soaps (micelles), and functional groups. 5. **Life Processes (6 Marks)**: Nephron structure & function, Double circulation in human heart, and Aerobic vs Anaerobic respiration pathways. 6. **Heredity (4 Marks)**: Mendel's monohybrid cross ($3:1$ phenotypic ratio) and sex determination in humans (XX and XY). 7. **Environment (4 Marks)**: 10% law of energy flow, Food chain vs food web, and Ozone layer depletion.